

Meet the Machines: Ollama, Hugging Face, and Critical Comparison

Overview

This lab focuses on the initial setup and exploration of Ollama and Hugging Face. Ollama is a free, open-source platform used to download, run, and manage large language models (LLMs) directly on a local computer, while Hugging Face is an open-source platform and central hub for discovering, sharing, training, and deploying AI and machine learning models.

The lab helps us understand how to read model cards and select suitable models based on available computing resources, memory, and processing power. We also compare different models using the same prompts and observe how their responses change when the audience, role, and format of the prompt change. In addition, we test the reliability of model outputs by verifying factual claims, checking citations and research papers, and reviewing generated code.

Finally, the lab highlights the importance of verifying AI-generated information, especially when using it for research or other important work.

Step 1: Install and verify Ollama

Field	Your answer
Model pulled	llama3.2
Time to first reply (seconds, rough)	less than a second
Screenshot filename	screenshots/llama_working.png

Step 2: Explore Hugging Face like an engineer

Field	Model 1	Model 2	Model 3
Full name and URL	meta-llama/Llama-3.2-3B-Instruct	mistralai/Mistral-7B-Instruct-v0.3	Qwen/Qwen2.5-1.5B-Instruct
Parameters (e.g. 3B, 7B)	3B	7B	2B
License (e.g. Apache 2.0, Llama, MIT)	Llama 3.2 Community License	Apache 2.0	Apache 2.0

Field	Model 1	Model 2	Model 3
Intended use (from card)	Commercial and research use; assistant-like chat, agentic applications, knowledge retrieval, summarization, mobile AI-powered writing assistants, query and prompt rewriting, and other natural language generation tasks.	It is an instruct fine-tuned version of the Mistral-7B-v0.3.	Instruction-following tasks, chat, long-text generation, structured data understanding, structured output generation, coding, mathematics, and multilingual applications.
Known limitations	Testing has not covered all possible scenarios, so Llama 3.2's outputs cannot be predicted in advance. It may produce inaccurate, biased, or otherwise objectionable responses. Developers should perform safety testing and tuning for their specific applications before deployment.	Does not have any moderation mechanisms. Working on ways to make the model finely respect guardrails, allowing for deployment in environments requiring moderated outputs.	Requires a recent version of Transformers. The model has a 32K-token context length and 8K-token generation limit.

Field	Model 1	Model 2	Model 3
Languages supported	English, German, French, Italian, Portuguese, Hindi, Spanish, Thai	Primarily English, multilingual capability is not stated.	Over 29 languages, including Chinese, English, French, Spanish, Portuguese, German, Italian, Russian, Japanese, Korean, Vietnamese, Thai, Arabic and more
Available on Ollama? (yes/no/check library)	Yes (llama3.2:3b)	Yes (mistral:7b-instruct)	Yes (qwen2.5:1.5b-instruct)

Questions

1. Which model would you not use in a commercial product? Quote the license line that supports your answer:

All 3 models can be used for commercial purposes. In the Llama intended use cases it clearly states Llama 3.2 is intended for commercial and research use in multiple languages (License Rights and Redistribution. Grant of Rights. You are granted a non-exclusive, worldwide, non-transferable and royalty-free limited license under Meta’s intellectual property or other rights owned by Meta embodied in the Llama Materials to use, reproduce, distribute, copy, create derivative works of, and make modifications to the Llama Materials.). While Mistral and Qwen have the Apache 2.0 licence which allows for unrestricted commercial use. (Subject to the terms and conditions of this License, each Contributor hereby grants to You a perpetual, worldwide, non-exclusive, no-charge, royalty-free, irrevocable copyright license to reproduce, prepare Derivative Works of, publicly display, publicly perform, sublicense, and distribute the Work and such Derivative Works in Source or Object form.)

2. Which model fits a student laptop best for local experiments? Why?:

Qwen2.5-1.5B-Instruct fits best because it has only 1.54B parameters and 986 MB in size, which makes it the smallest of the three models. A smaller model generally requires fewer computing resources, memory and processing power needed to run or train it.

3. What is one risk mentioned in a model card that you had not thought about before?:

Llama 3.2’s potential outputs cannot be predicted in advance and it may produce

inaccurate, biased, or otherwise objectionable responses. I'm wondering how reliable that makes it if it's behaviour is unpredictable. It also mentioned the model is supposed to be deployed as part of an overall AI system with additional safety guardrails as required and it is the developers responsibility to ensure that. Is that what people see as the AI model going rogue?

Step 3: The same prompt experiment

Run	System	Model name (exact tag)
A	Ollama	llama3.2
B	Ollama (different model)	mistral:7b-instruct
C	Third system	qwen2.5:1.5b-instruct

Explain overfitting in machine learning using a real-world analogy. Include one sentence on how to reduce it. Keep the answer under 150 words.

Run A = 3 secs Run B = 4 secs Run C = 1 sec

Criterion	Run A	Run B	Run C
Accuracy (is the ML idea correct?)	5	5	3
Clarity (would a classmate understand?)	5	5	4
Completeness (analogy + reduction tip?)	5	5	4
Respects ~150 word limit	5	5	5
Overall	20	20	16

Best model(Run B) - mistral:7b-instruct - the clarity and accuracy with how it explained the concept. It explained the poor performance on unseen data clearly and gave an explanation on how increasing the training data helps with overfitting.

Step 4: Audience shift experiment - using the best model

Prompt	What changed besides vocabulary?	New factual claim?	Trust it?
Child	Simplifies the explanation but still sounds high level for a 12 year old.	No new factual claim.	Yes

Prompt	What changed besides vocabulary?	New factual claim?	Trust it?
Student	Adds specific techniques for avoiding overfitting like cross-validation, regularization, and simpler models.	Yes. These are factual claims about techniques used to reduce overfitting.	Yes
Researcher	Adds the bias-variance tradeoff, explains bias, variance	Yes. It introduces additional factual technical claims.	Yes

For the child response (outputs/step4_child.txt) I will change “In machine learning, a model can learn too much from the training data and perform poorly on new, unseen data. It’s important to find a balance so the model can learn well but not too much” to “The goal is to focus on the general features instead of highlighting every little detail.” which follows the drawing analogy and substitutes all the technical words for everyday language for a 12 year old

Step 5: Reliability and hallucination hunt

Verifiable fact test

Model output	Your verification source	Verdict (correct / incorrect / ambiguous)
Canberra	Google	correct

Citation trap

Title (as model gave it)	Exists?	Authors match?	Notes
Attention is All You Need	Yes	All but 1	S. Tenenbaum is not an author on this paper

Title (as model gave it)	Exists?	Authors match?	Notes
Transformer-XL: Longer, Faster, Better Models for Long Sequences	Yes but the title is incorrect	No	Title is Transformer-XL: Attentive Language Models Beyond a Fixed-Length Context and the authors are Zihang Dai, Zhilin Yang, Yiming Yang, Jaime Carbonell, Quoc V. Le, and Ruslan Salakhutdinov
RoBERTa: A Robustly Optimized BERT Pretraining Approach	Yes	All but 1	The correct name is Veselin Stoyanov not Mark Neumann

Code sanity check

Check	Result
Runs without syntax error?	Yes
Formula looks correct?	Yes
Any invented NumPy functions?	No

Step 6: Toolchain reflection

1. Ollama:

What worked well?: I could test models such as Llama, Mistral, and Qwen locally and run comparisons on their outputs easily. What frustrated you?: I couldn't immediately find the models I was reviewing on Hugging face on Ollama. Especially when trying to find the exact versions.

2. Hugging Face: What would you check before using a model in a client project?

I would check its license to make sure commercial use is allowed. I would also check the model's intended use, limitations, supported languages, and parameter size. I would review the model card to understand how the model was trained

and any known risks or limitations. I would also test the model with relevant data to see whether its performance is suitable for the project.

3. Modal (conceptual): Describe one task from this lab that would not run well on your laptop but would need a cloud GPU.

Running or fine-tuning a large language model. Larger models require a lot of memory and computing power, especially when training or fine-tuning them. A cloud GPU would provide the processing power and memory needed to run the model efficiently. Allowing the task to complete much faster than it would on my laptop.

4. Trust:

Under what conditions would you trust output from your local Ollama model without verification?: When it is being used for simple tasks such as brainstorming, rewriting text, explaining basic concepts that I already understand or if I asked the model for an opinion. For important factual information, I would verify the output against reliable sources. Under what conditions would you never?: For medical, legal, financial, any information where specific facts, statistics, citations, claims are provided or for any other high-stakes decisions.

Defrief

The models (Llama, Mistral) when answering general knowledge questions and definitions of data science concepts were more factual, except Qwen that was confident in explaining techniques for reducing overfitting. But when asked to cite research papers Mistral started to hallucinate with the names of authors and titles of the research papers. When working with AI-generated text about content I am not sure of I will always verify because I could have made reference to research papers that didn't exist in important school work or research of my own.